

Lecture 5: Lagrange Interpolation

CSE 330 - Numerical Methods (Class Notes)

1 Derivation of the Method

In this method, we use a new basis $\{\ell_k(x)\}$ for $k = 0, 1, 2, \dots, n$, called the **Lagrange basis**, to express the n -degree polynomial $p_n(x)$ as

$$p_n(x) = \sum_{k=0}^n \ell_k(x) f(x_k) = \ell_0(x)f(x_0) + \ell_1(x)f(x_1) + \ell_2(x)f(x_2) + \dots + \ell_n(x)f(x_n),$$

where the $f(x_k)$'s are the function values at the nodal points. In the following we explore the properties of the Lagrange bases $\ell_k(x)$ and their algebraic expressions.

The case $n = 1$

To start with, we take $n = 1$ and write $p_1(x)$ in the natural (Taylor) basis,

$$p_1(x) = a_0 + a_1x. \tag{12}$$

For $n = 1$ we have two nodes, x_0 and x_1 . Applying the interpolation conditions $p_1(x_i) = f(x_i)$ at the nodes gives two equations:

$$\begin{aligned} p_1(x_0) = f(x_0) &\implies a_0 + a_1x_0 = f(x_0), \\ p_1(x_1) = f(x_1) &\implies a_0 + a_1x_1 = f(x_1). \end{aligned} \tag{13}$$

Solving Eq.(13) for a_0 and a_1 :

$$a_0 = \frac{x_1f(x_0) - x_0f(x_1)}{x_1 - x_0}, \quad a_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}.$$

Substituting these back, we can express $p_1(x)$ as

$$\begin{aligned} p_1(x) = a_0 + a_1x &= \frac{x_1f(x_0) - x_0f(x_1)}{x_1 - x_0} + \frac{f(x_1) - f(x_0)}{x_1 - x_0}x \\ &= \frac{x - x_1}{x_0 - x_1}f(x_0) + \frac{x - x_0}{x_1 - x_0}f(x_1). \end{aligned}$$

We now define the two Lagrange basis elements for $n = 1$ as

$$\ell_0(x) \equiv \frac{x - x_1}{x_0 - x_1} \quad \text{and} \quad \ell_1(x) \equiv \frac{x - x_0}{x_1 - x_0}. \tag{14}$$

These elements satisfy

$$\ell_0(x_0) = 1, \quad \ell_0(x_1) = 0, \quad \ell_1(x_0) = 0, \quad \ell_1(x_1) = 1,$$

which can be summarized by the single relation

$$\ell_k(x_j) = \begin{cases} 1, & k = j, \\ 0, & k \neq j; \end{cases} \quad j, k = 0, 1.$$

Note that for $n = 1$ there are exactly two basis elements (equal to the number of nodes), and each element contains only one ratio (the same as the value of n).

Generalizing to arbitrary n

For $(n + 1)$ nodes $x_0, x_1, \dots, x_n$ we define the Lagrange basis $\ell_k(x)$, $k = 0, 1, \dots, n$, so that each element satisfies

$$\ell_k(x_j) = \begin{cases} 1, & k = j, \\ 0, & k \neq j; \end{cases} \quad j, k = 0, 1, 2, \dots, n. \quad (15)$$

Treating $\ell_k(x)$ as a function of x , Eq.(15) says $\ell_k(x_j) = 0$ for all $j \neq k$. So $x = x_0, x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_n$ are the *roots* of $\ell_k(x)$. By the properties of algebra we can therefore write

$$\begin{aligned} \ell_k(x) &\propto (x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n), \\ \implies \ell_k(x) &= B(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n). \end{aligned}$$

Now use $\ell_k(x_k) = 1$ from Eq.(15) to determine the constant B :

$$\begin{aligned} \ell_k(x_k) &= B(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n) = 1, \\ \implies B &= \frac{1}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)}. \end{aligned}$$

Substituting B back gives the general Lagrange basis and interpolating polynomial:

Lagrange basis:

$$\ell_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n \frac{x - x_j}{x_k - x_j}.$$

Lagrange expression:

$$p_n(x) = \sum_{k=0}^n \ell_k(x) f(x_k).$$

By Eq.(15), $p_n(x_j) = \sum_k \ell_k(x_j) f(x_k) = f(x_j)$, so the polynomial passes exactly through every data point.

For three points ($n = 2$) this reads

$$\begin{aligned} p_2(x) &= \ell_0(x)f(x_0) + \ell_1(x)f(x_1) + \ell_2(x)f(x_2), \\ \ell_0(x) &= \frac{x - x_1}{x_0 - x_1} \cdot \frac{x - x_2}{x_0 - x_2}, \quad \ell_1(x) = \frac{x - x_0}{x_1 - x_0} \cdot \frac{x - x_2}{x_1 - x_2}, \quad \ell_2(x) = \frac{x - x_0}{x_2 - x_0} \cdot \frac{x - x_1}{x_2 - x_1}. \end{aligned}$$

2 Example 1 - Age vs. Salary

	x (age)	$f(x)$ (salary)
x_0	5	100
x_1	10	200
x_2	20	300

Build the basis polynomials

$$\ell_0(x) = \frac{x-10}{5-10} \cdot \frac{x-20}{5-20} = \frac{x-10}{-5} \cdot \frac{x-20}{-15} = \frac{(x-10)(x-20)}{75},$$

$$\ell_1(x) = \frac{x-5}{10-5} \cdot \frac{x-20}{10-20} = \frac{x-5}{5} \cdot \frac{x-20}{-10} = \frac{(x-5)(x-20)}{-50},$$

$$\ell_2(x) = \frac{x-5}{20-5} \cdot \frac{x-10}{20-10} = \frac{x-5}{15} \cdot \frac{x-10}{10} = \frac{(x-5)(x-10)}{150}.$$

Watch the sign on ℓ_1 . Its denominator is $(x_1 - x_0)(x_1 - x_2) = (10 - 5)(10 - 20) = (5)(-10) = -50$, so $\ell_1(x) = \frac{(x-5)(x-20)}{-50}$ (not $+50$).

Put the basis values into $p_2(x)$

$$p_2(x) = \frac{(x-10)(x-20)}{75} \times 100 + \frac{(x-5)(x-20)}{-50} \times 200 + \frac{(x-5)(x-10)}{150} \times 300.$$

Evaluate $p_2(30)$

$$\ell_0(30) = \frac{(30-10)(30-20)}{75} = \frac{(20)(10)}{75} = \frac{200}{75} = 2.6667,$$

$$\ell_1(30) = \frac{(30-5)(30-20)}{-50} = \frac{(25)(10)}{-50} = \frac{250}{-50} = -5,$$

$$\ell_2(30) = \frac{(30-5)(30-10)}{150} = \frac{(25)(20)}{150} = \frac{500}{150} = 3.3333.$$

$$\begin{aligned} p_2(30) &= (2.6667)(100) + (-5)(200) + (3.3333)(300) \\ &= 266.67 - 1000 + 1000 \\ &= \boxed{266.67}. \end{aligned}$$

Remark. If the ℓ_1 denominator is taken as $+50$ (dropping the minus sign), the middle term flips to $+1000$ and one gets 2266.67 . The sign-correct value is $p_2(30) = 266.67$.

Limitation. For a *new* x -value we must recompute *all* the basis polynomials over again (every ℓ_k depends on all the nodes). A more incremental solution is **Newton's** divided-difference method.

3 Example 2 - $f(x) = \sin(x)$

Take the nodes

$$x_0 = 0, \quad x_1 = \frac{\pi}{2}, \quad x_2 = \frac{\pi}{4},$$

with

$$f(x_0) = \sin 0 = 0, \quad f(x_1) = \sin \frac{\pi}{2} = 1, \quad f(x_2) = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}.$$

Build the basis polynomials

$$\ell_0(x) = \frac{x - \frac{\pi}{2}}{0 - \frac{\pi}{2}} \cdot \frac{x - \frac{\pi}{4}}{0 - \frac{\pi}{4}} = \frac{(x - \frac{\pi}{2})(x - \frac{\pi}{4})}{\frac{\pi^2}{8}} = \frac{8}{\pi^2} \left(x - \frac{\pi}{2}\right) \left(x - \frac{\pi}{4}\right),$$

$$\ell_1(x) = \frac{x - 0}{\frac{\pi}{2} - 0} \cdot \frac{x - \frac{\pi}{4}}{\frac{\pi}{2} - \frac{\pi}{4}} = \frac{x(x - \frac{\pi}{4})}{\frac{\pi^2}{8}} = \frac{8}{\pi^2} x \left(x - \frac{\pi}{4}\right),$$

$$\ell_2(x) = \frac{x - 0}{\frac{\pi}{4} - 0} \cdot \frac{x - \frac{\pi}{2}}{\frac{\pi}{4} - \frac{\pi}{2}} = \frac{x(x - \frac{\pi}{2})}{-\frac{\pi^2}{16}} = -\frac{16}{\pi^2} x \left(x - \frac{\pi}{2}\right).$$

Assemble $p_2(x)$

Since $f(x_0) = 0$, the ℓ_0 term drops out:

$$\begin{aligned} p_2(x) &= \ell_0(x) \cdot 0 + \ell_1(x) \cdot 1 + \ell_2(x) \cdot \frac{1}{\sqrt{2}} \\ &= \frac{8}{\pi^2} x \left(x - \frac{\pi}{4}\right) - \frac{16}{\pi^2} x \left(x - \frac{\pi}{2}\right) \cdot \frac{1}{\sqrt{2}} \\ &= \frac{8}{\pi^2} x \left(x - \frac{\pi}{4}\right) - \frac{8\sqrt{2}}{\pi^2} x \left(x - \frac{\pi}{2}\right). \end{aligned}$$

(Check: $p_2(0) = 0$, $p_2(\frac{\pi}{2}) = 1$, $p_2(\frac{\pi}{4}) = \frac{1}{\sqrt{2}}$, matching the data - as guaranteed by the Kronecker-delta property, Eq.(15).)

4 Choosing Local Nodes - Age vs. Score

age	score
25	100
30	150
40	250
50	280
65	290

To estimate the score at age 45, i.e. $p_2(45)$, we pick the three nearest nodes that bracket 45:

$$x_0 = 30 (150), \quad x_1 = 40 (250), \quad x_2 = 50 (280).$$

$$\ell_0(45) = \frac{(45 - 40)(45 - 50)}{(30 - 40)(30 - 50)} = \frac{(5)(-5)}{(-10)(-20)} = -0.125,$$

$$\ell_1(45) = \frac{(45 - 30)(45 - 50)}{(40 - 30)(40 - 50)} = \frac{(15)(-5)}{(10)(-10)} = 0.75,$$

$$\ell_2(45) = \frac{(45 - 30)(45 - 40)}{(50 - 30)(50 - 40)} = \frac{(15)(5)}{(20)(10)} = 0.375.$$

$$p_2(45) = (-0.125)(150) + (0.75)(250) + (0.375)(280) = -18.75 + 187.5 + 105 = \boxed{273.75}.$$

Limitation (extra term). Choosing different nodes (or adding one) means *all* basis polynomials change and must be rebuilt - the “extra term” problem. Newton’s divided-difference method avoids this by extending the polynomial one term at a time.